

Bracket Arbitrage Problems

16 teams play a single-elimination bracket: A plays B, C plays D, and so on; the winner of A–B plays the winner of C–D, and the same for all teams (the most trivial format). You can buy any team at its listed price; a team pays out according to the level it reaches. Prize lines read: Winner / Runner-up / Semi-finalist (3rd–4th) / Quarter-finalist (5th–8th) / First-round loser. You cannot sell a team and cannot buy more than one of a team, except where a problem says otherwise.

Problem 1. Prizes: 1500 / 900 / 700 / 400 / 0

A 150	B 200	C 250	D 300
E 400	F 500	G 350	H 400
I 200	J 250	K 300	L 350
M 150	N 150	O 250	P 300

(a) Find all possible arbitrage strategies. (b) Which guarantees the most profit? Which guarantees the most per unit spent?

Problem 2. Prizes: 1500 / 900 / 700 / 400 / 0

A 250	B 350	C 300	D 400
E 350	F 450	G 500	H 600
I 100	J 150	K 250	L 250
M 250	N 300	O 300	P 350

(a) Find all possible arbitrage strategies. (b) You have exactly 750: what is the maximum profit you can guarantee? (c) You may borrow above 750 at a flat 20% fee. Worth it? At what fee does the answer flip?

Problem 3. Prizes: 1500 / 900 / 700 / 400 / 0

A 200	B 300	C 250	D 250
E 500	F 600	G 700	H 600
I 250	J 300	K 350	L 300
M 150	N 250	O 400	P 350

(a) Find all possible arbitrage strategies. (b) Is buying all 16 teams an arbitrage?

Problem 4. Prizes: 2000 / 1200 / 800 / 500 / 150

A 350	B 350	C 350	D 400
E 400	F 500	G 1100	H 1000
I 450	J 500	K 500	L 500
M 500	N 120	O 500	P 550

(a) Find all possible arbitrage strategies. (b) Teams I–P bought together: arbitrage or not, and by exactly how much?

Problem 5. Prizes: 1500 / 900 / 700 / 400 / 0

Selling allowed: you may sell any team at its price minus 50; you pay a sold team's full prize at the end.

A 150	B 200	C 300	D 400
E 250	F 350	G 950	H 850
I 250	J 300	K 250	L 250
M 200	N 250	O 250	P 250

(a) Find all possible arbitrage strategies. (b) What is the maximum profit you can guarantee, and in which exact final result do you earn the least?

Problem 6. Prizes: 2500 / 1200 / 600 / 300 / 0

Selling allowed: you may sell any team at its price minus 50; you pay a sold team's full prize at the end.

A 130	B 150	C 220	D 280
E 320	F 380	G 1600	H 1500
I 100	J 150	K 230	L 270
M 250	N 300	O 280	P 320

(a) Find all possible arbitrage strategies. (b) Maximum guaranteed profit?

Problem 7. Prizes: as in Problem 3 (same teams and prices)

New rule: after every round, the surviving teams are randomly re-paired.

(a) Which of your Problem 3 strategies still work? (b) What is the worst-case payout of owning k full first-round pairs, for $k = 1$ to 8? (c) What is the cheapest way to guarantee one of your teams plays a semi-final, and does it make money?

Problem 8. Prizes: as in Problem 5 (same teams, prices, and selling rule)

The re-pairing rule of Problem 7 applies here too.

(a) Does selling E, F, G, H together still guarantee a profit? (b) Maximum guaranteed profit?

Problem 9. Prizes: as in Problem 2 (same teams and prices)

Each team has a second unit available at its price plus 50; units pay prizes independently; at most two units per team.

Maximum guaranteed profit, and how many units of each team do you buy?

Problem 10. Prizes: Winner 1000, Runner-up 600, Third 450, Fourth 250, others 0

Eight teams only. A plays B, C plays D, E plays F, G plays H; the A–B and C–D winners meet in one semi-final, the E–F and G–H winners in the other; the two semi-final losers play a third-place match.

A 100	B 150	C 400	D 500
E 150	F 200	G 400	H 450

(a) Find all possible arbitrage strategies. (b) After the quarter-finals, each surviving team is re-offered at 1.5 times its price; you may buy now and/or then, committing your full plan in advance. Maximum guaranteed profit now? (c) At what multiplier does your (b) plan stop working?

Problem 11. Prizes: 1500 / 900 / 700 / 400 / 0

(a) You set all 16 prices yourself. What is the smallest possible total of the 16 prices such that no buying strategy is an arbitrage? Give prices achieving it. (b) You also quote selling at price minus 50. What is the largest possible total such that no selling strategy is an arbitrage?